MAKERERE UNIVERSITY

College of Computing and Information Sciences

School of Computing and Informatics Technology

Department of Networks

BSE4104 - Emerging Trends in Software Engineering

AI Native and Agentic Engineering Capstone Project

GROUP J EVENING

WEEK 1: PROBLEM FRAMING AND AI NATIVE REQUIREMENTS

SME PROCUREMENT PREPARATION AGENT

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TESTABLE USER STORIES AND ACCEPTANCE CRITERIA

ID	USER STORY	ACCEPTANCE CRITERIA
US-1	As a procurement officer, I want the system to identify items whose current stock has fallen to or below the reorder point so that I can act before a stock out occurs.	1. When current stock <= reorder point, the item is flagged as 'Reorder Required'. 2. The flag displays item name or ID, current stock, reorder point and recommended quantity. 3. Items with stock above the reorder point are never flagged. 4. Zero or negative stock values are handled without system error and are correctly flagged. 5. Response time for a single item or full inventory check is <= 3 seconds on the synthetic dataset.
US-2	As a procurement officer, I want a recommended reorder quantity based on defined rules so that I order an appropriate amount.	1. The recommended quantity is calculated by deterministic code using the agreed formula. 2. The AI may explain the recommendation in natural language but must not alter the calculated number. 3. When usage data is missing, the system falls back to the static reorder quantity and explicitly states the limitation. 4. The recommended quantity is always a positive integer. 5. The calculation is fully traceable.
US-3	As a procurement officer, I want the system to retrieve and compare all valid supplier quotations for a selected item so that I can see price and delivery differences.	1. All non expired quotations for the item are returned with, supplier name, unit price, total cost for recommended quantity, lead time and validity date. 2. Quotations are ranked by unit price by default with the lowest first. 3. Expired quotations are either excluded or clearly marked 'Expired'. 4. When no valid quotations exist, the system returns an explicit 'No valid

		quotations found' message and does not invent data. 5. Every quotation shown is linked to its source record ID for auditability.
US-4	As a procurement officer, I want the system to generate a complete draft purchase requisition so that I save time on manual paperwork.	1. The draft contains at minimum, item ID, name, quantity, estimated total cost, top 2 - 3 supplier options with prices, justification text and generation date. 2. All numerical values for quantity, prices and totals are produced by deterministic numbers. 3. The user can request regeneration after changing quantity or preferred supplier options. 4. The draft response lists the inventory and quotation records used.
US5	As a procurement officer, I want every draft requisition to be routed for explicit human approval before any further action so that no purchase commitment occurs without human control.	1. No requisition can leave 'Draft' status without a recorded human decision. 2. The decision records approver identity or simulated ID, timestamp, decision and optional comments. 3. 'Reject' or 'Request Changes' returns the case to the requester with comments visible. 4. The agent is technically blocked from calling any send or commit or order tool until approval is recorded. 5. Attempted bypass of the approval gate is logged as a policy violation and prevented.
US-6	As a user, I want the system to remember the status and content of an active requisition document so that I can resume work later without starting over.	1. Opening an existing case restores the latest draft, quotations used, current status and decision history. 2. Only justified data is stored. 3. A user can explicitly close or archive a case where archived cases are removed from the active agent context. 4. Memory never silently overrides supplier selection or approval decisions. 5. Retention and deletion rules are documented and enforceable.
US-7	As a user, I want every recommendation and draft to	1. Every AI generated explanation or draft includes the source record IDs.

	show the exact inventory and quotation records used so that I can trust and audit the output.	2. If sufficient grounded data is unavailable, the system refuses to answer and states 'Insufficient data'. 3. Full execution traces are logged for every run. 4. Traces are inspectable during demonstration and evaluation.
US-8	As a user, I want the system to handle missing inventory data and failures without inventing information so that I am never misled.	1. If item is unknown, systems displays a clear message 'Item not found in inventory'. 2. The agent never proceeds to draft creation or approval routing when required data is absent. 3. All failure cases are covered in the evaluation scenario set.
US-9	As a system owner, I want the agent prevented from performing any high impact or prohibited action so that the business cannot be financially committed by the AI.	1. Any attempt to call prohibited actions like place order, send to supplier, commit budget or select final supplier autonomously is blocked and logged. 2. High impact actions remain behind the explicit human approval gate at all times. 3. Policy violation attempts appear in the Failure Catalogue and evaluation results.
US-10	As a development team member, I want every agent run to produce structured and inspectable traces so that we can evaluate reliability, diagnose failures and demonstrate control.	1. Traces are stored in a form that can be shown during the live demonstration and included in the evidence pack. 2. Key metrics such as task completion, safety violations and tool selection accuracy can be derived from the traces.
US-11	As a procurement officer, I want the system to remember my previously preferred suppliers for an item so that future drafts can highlight those suppliers without me reselecting them every time.	1. A user can mark one or more suppliers as 'Preferred' for a specific item after reviewing quotations. 2. On subsequent runs for the same item, preferred suppliers are highlighted at the top of the comparison list. 3. Preference is stored only after explicit user action and can be removed by the user. 4. Preference never automatically selects a supplier or bypasses the human approval gate.

		5. Preference data is part of the justified persistent memory and is visible in the case record.
US-12	As a procurement officer, I want the system to update the status and provide a clear confirmation after I approve or reject a draft so that I have an immediate record of the decision and the case is correctly closed or returned.	1. After an ‘Approve’ decision, the case status changes to ‘Approved' and a confirmation message is shown. 2. After a ‘Reject’ or ‘Request Changes’ decision, the status returns to ‘Draft needs review’ and the comments are attached. 3. The decision is permanently recorded in the case history. 4. The agent loop terminates cleanly after the decision and no further autonomous actions are taken. 5. The updated status and confirmation are visible both in the UI and in the execution trace.